

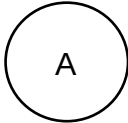
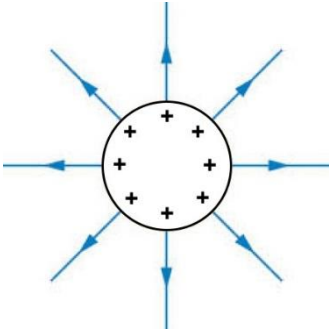
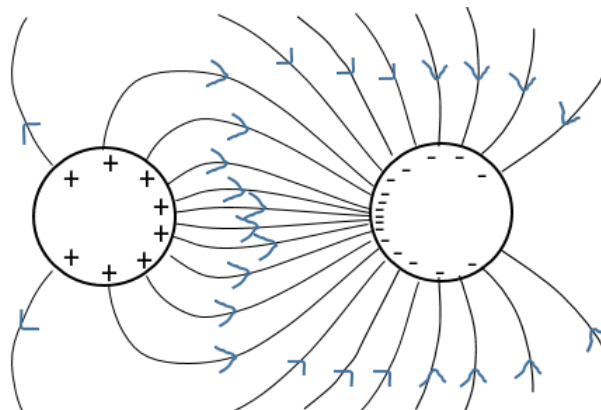
8	(a)	Explain what is meant by an <i>electric field</i> .	
			[1]
		It is the <u>region of space</u> where a <u>charged particle</u> experiences an <u>electric force</u> .	1
	(b)	The charges on an isolated metal sphere are uniformly distributed on its surface. Fig.8.1 shows a positively charged metal sphere A. On Fig.8.1, draw the charge distribution on the sphere and the electric field around it.	
			
		Fig. 8.1	[1]
		Solution: 	
		• <u>equal spacing</u> of charge on the <u>surface</u> of sphere, and, • <u>radial</u> field acting <u>outwards</u> • field lines drawn <u>perpendicular</u> to the surface of the metal sphere	1
	(c)	A negatively charged metal sphere B is brought close to the positively charged metal sphere A as shown in Fig.8.2. The charge on metal sphere B is twice that of the charge on metal sphere A. On Fig. 8.2, draw the charge distribution on the spheres and the electric field around the spheres.	



Fig. 8.2

[3]

Solution:



Mark scheme:

- Distribution of charges: correct distribution, charges drawn on the surfaces, twice the number of '-' on B compared to the number of '+' on A. **1**
- Field pattern: correct shape & variation of density of field lines, number of field lines around B twice that around A. **1**
- Direction of field lines, including field lines drawn perpendicular to the surface of each metal sphere. **1**

OR

- Distribution of charges: correct distribution, charges drawn on the surfaces, twice the number of '-' on B compared to the number of '+' on A. **[1]**
- Field pattern & Direction of field lines **[1]**
- Details: number of line around each sphere, lines drawn perpendicular to the surface of the spheres. **[1]**

- (d) Point P is at a distance x from the centre of sphere A along the line joining the centres of the two spheres as shown in Fig.8.3. The radius of A and B is 15 mm and the distance between the centres of the spheres is 80 mm.

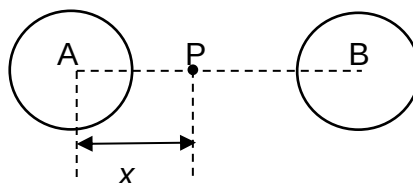


Fig. 8.3

The variation with x of the electric potential V at P is shown in Fig. 8.4.

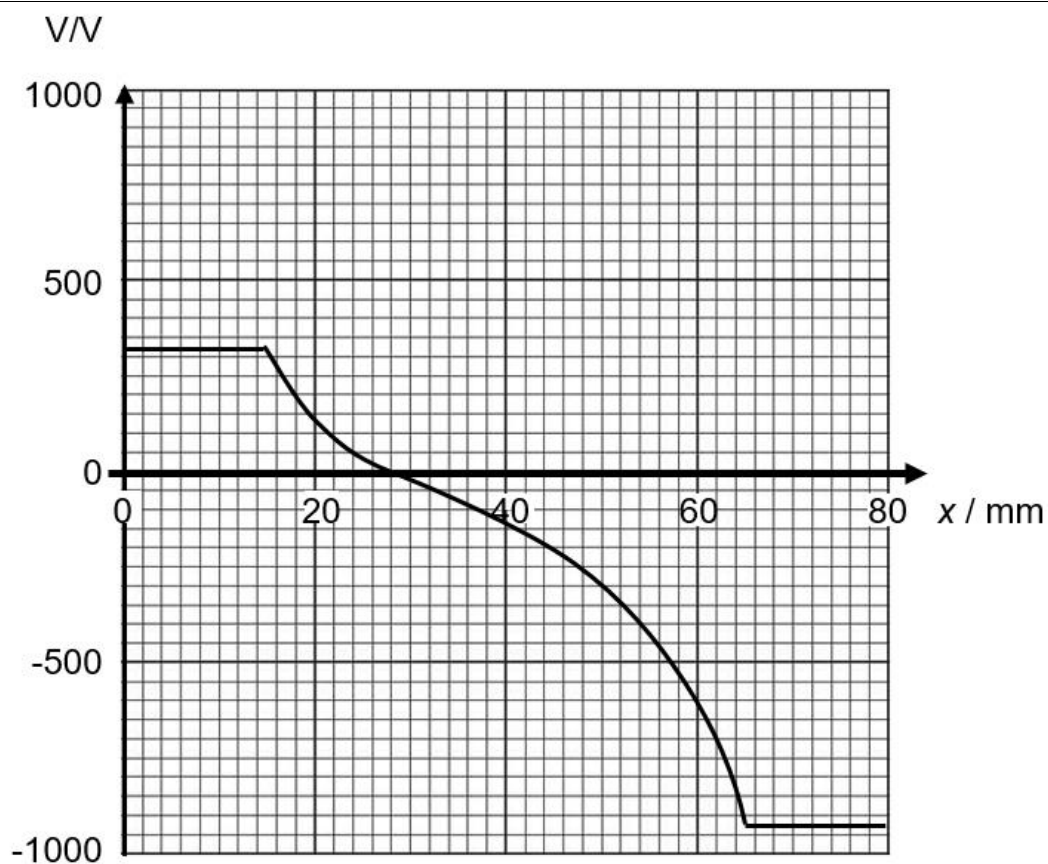


Fig. 8.4

- (i) Determine the magnitude of the electric field strength at P where $x = 40$ mm.

magnitude of the electric field strength = V m^{-1} [2]

$$\begin{aligned}
 |E| &= |dV / dx| \\
 &= \text{gradient of } V\text{-}x \text{ graph at } x = 40 \text{ mm} \\
 &= 350 - (-600) / 80 \times 10^{-3} \\
 &= 1.19 \times 10^4 \text{ V m}^{-1}
 \end{aligned}$$

1

1

		(ii)	An electron is initially at rest at point P where $x = 40$ mm.	
		1.	Describe and explain the motion of the electron as it travels 20 mm along the line joining the centres of the spheres.	
				
				
				[3]
			$a = \frac{F_{net}}{m} = \frac{F_E}{m} = \frac{eE}{m}$ where $E = -\frac{dV}{dx}$ <u>Gradient of the V-x graph is proportional to the electric field strength which is proportional to acceleration.</u> <u>When gradient decreases from 40 mm to 28 mm, acceleration decreases.</u> <u>→ The electron moves to the left towards A, speeds up at a decreasing rate.</u> <u>When gradient increases from 28 mm to 20 mm, acceleration increases.</u> <u>→ The electron continues moving to the left towards A, speeds up at an increasing rate.</u>	1 1 1
		2.	Determine the speed of the electron when it has travelled 20 mm along the line joining the centres of the spheres.	
				
			speed of electron =m s ⁻¹	[3]
			moving 20 mm towards A, the electron moves from $x = 40$ mm to $x = 20$ mm, increase in potential, $\Delta V = 125 - (-150) = 275$ V <i>[Also accept V_{40mm} between -125 V to -150 V; V_{20mm} between 125 V to 150 V]</i> decrease in potential energy, $\Delta U = q \cdot \Delta V = 1.60 \times 10^{-19} \times 275 = 4.40 \times 10^{-17}$ J By conservation of energy, increase in KE = decrease in electric potential energy final KE – initial KE = ΔU $\frac{1}{2} m v^2 - 0 = 4.40 \times 10^{-17}$ $\frac{1}{2} (9.11 \times 10^{-31}) v^2 = 4.40 \times 10^{-17}$ $v = 9.83 \times 10^6$ m s⁻¹	1 1 1

- (e) An electron is projected along the line XY into a region of uniform electric field between two charged parallel plates of length 20.0 cm separated by 8.0 cm, as shown in the Fig 8.5. The potential difference between the two plates is 200 V. Between the plates, the electron travels along a curved path and exit the region between the plates at point Q which is 3.0 cm from the line XY.

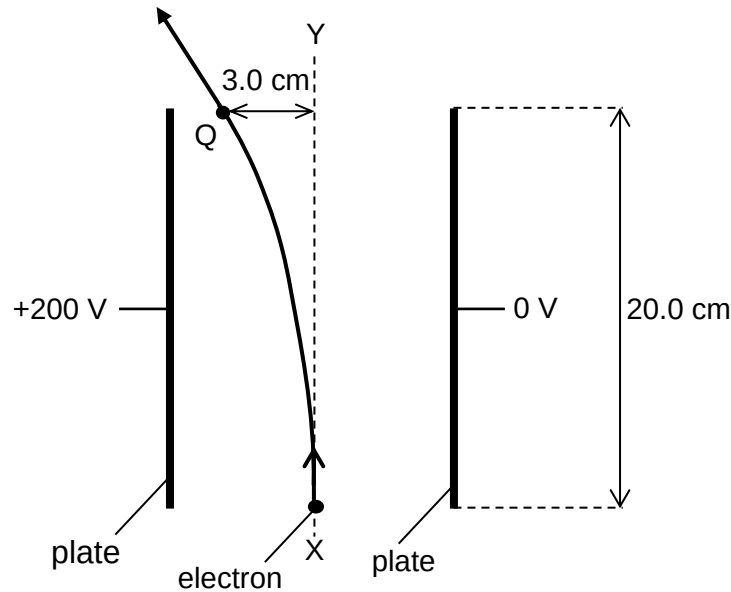


Fig. 8.5

- (i) Calculate the electric field strength between the two plates.

electric field strength = V m⁻¹ [1]

$E = V/d = 200 / 0.080 = 2500 \text{ V m}^{-1}$ **1**

- (ii) Calculate the initial speed of the electron projected into the electric field.

speed = m s⁻¹ [4]

consider motion perpendicular to XY

Electric force $F = qE = 1.60 \times 10^{-19} \times 2500$

acceleration $= F / m = 1.60 \times 10^{-19} \times 2500 / 9.11 \times 10^{-31} = 4.396 \times 10^{14}$ **1**

Using $s = ut + \frac{1}{2}at^2$

$0.030 = 0 + \frac{1}{2}(4.396 \times 10^{14})t^2$ (2) **1**

$t = 1.168 \times 10^{-8} \text{ s}$

consider motion along XY

Using $s = ut + \frac{1}{2}at^2$

$0.200 = u(1.168 \times 10^{-8}) + 0$

$u = 1.712 \times 10^7 = 1.7 \times 10^7 \text{ m s}^{-1}$ **1**

- (iii) A proton is now projected into the same electric field and with the same velocity as that of the electron.

Explain why the deflection of the proton is much lesser compared to the deflection of the electron.

.....

				[2]
			A proton has the same magnitude of charge as the electron, hence it will experience the same magnitude of electric force as that on the electron. A proton has a mass about <u>1800 times</u> that of the electron (or <u>much more massive than</u> that of the electron), hence it will experience an acceleration <u>1800 times less than</u> that on the electron (or <u>much smaller acceleration</u>). Time spent between the plates is the same for both the proton and electron, hence the deflection will be much lesser and the path is much less curved.	1 1

9	(a)	A satellite orbits the Earth of mass M in a circular path of radius r
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